ATO / ANALOG REFERENCE

5 V / ARTIFICIAL SIGNALS / ELECTRICAL TEST LOADS ONLY

RECORDING + BUFFERED MIDPOINT

PRECISION SINK + POLARITY SWITCH

J1
+5V 1
GND 2
1
2
5V INPUT

J2
SIG_IN 1
GND 2
MID 3
1
2
3
ARTIFICIAL IN

J3
REC_OUT 1
GND 2
MID 3
1
2
3
ANALOG OUT

PWR_FLAG
+5V

PWR_FLAG
GND

U1
+5V 8
REC_IN 3
REC_FB 2
MID_RAW 5
MID 6
GND 4
V+
IN_A+
IN_A−
IN_B+
IN_B−
V−
OUT_A 1 REC_AMP
OUT_B 7 MID
OPA2333AID

U2
+5V 8
I_REF_RAW 3
I_REF 2
I_REF 5
I_SENSE 6
GND 4
V+
IN_A+
IN_A−
IN_B+
IN_B−
V−
OUT_A 1 I_REF
OUT_B 7 GATE_DRV
OPA2333AID

Q1
GATE 1
G
D 3 SINK
S 2 I_SENSE
2N7002

Q2
INHIBIT 1
G
D 3 +5V
S 2 SINK
2N7002

U3
+5V 16
INHIBIT 6
PHASE 11
PHASE 10
GND 9
GND 7
GND 8
VDD
INH
S_A
S_B
S_C
VEE
VSS
A0 12 +5V
A1 13 SINK
A 14 SW_A
B0 2 SINK
B1 1 +5V
B 15 SW_B
C0 5 GND
C1 3 GND
C 4 GND
74HC4053D

J4
PHASE 1
INHIBIT 2
+5V 3
GND 4
1
2
3
4
5V LOGIC

J5
DUT_A 1
DUT_B 2
1
2
DUMMY LOAD

PASSIVES / net labels define connectivity

R1 SIG_IN 1 | 1 2 | 2 AC_IN — 10k
C1 AC_IN 1 | 1 2 | 2 REC_IN — 1u
R2 REC_IN 1 | 1 2 | 2 MID — 1M
R3 REC_FB 1 | 1 2 | 2 MID — 1k
R4 REC_AMP 1 | 1 2 | 2 REC_FB — 20k
R5 REC_AMP 1 | 1 2 | 2 REC_OUT — 1k
C2 REC_OUT 1 | 1 2 | 2 GND — 47n

R6 +5V 1 | 1 2 | 2 MID_RAW — 10k
R7 MID_RAW 1 | 1 2 | 2 GND — 10k
C3 MID_RAW 1 | 1 2 | 2 GND — 10u
C5 +5V 1 | 1 2 | 2 GND — 100n
C8 +5V 1 | 1 2 | 2 GND — 10u
R8 +5V 1 | 1 2 | 2 I_REF_RAW — 49.9k
R9 I_REF_RAW 1 | 1 2 | 2 GND — 1.02k

C4 I_REF_RAW 1 | 2 GND — 100n
R10 I_SENSE 1 | 1 2 | 2 GND — 1k
R11 GATE_DRV 1 | 1 2 | 2 GATE — 1k
C6 +5V 1 | 1 2 | 2 GND — 100n
R12 INHIBIT 1 | 1 2 | 2 +5V — 100k
R13 PHASE 1 | 1 2 | 2 GND — 100k
C7 +5V 1 | 1 2 | 2 GND — 100n

R14 SW_A 1 | 1 2 | 2 DUT_A — 12k
R15 SW_B 1 | 1 2 | 2 DUT_B — 12k

OPA2333: OUTA=1, INA−=2, INA+=3, V−=4, INB+=5, INB−=6, OUTB=7, V+=8

PHASE 0: A=+5V, B=sink. PHASE 1: A=sink, B=+5V. INHIBIT 1: isolated.

External 5 V sequencer: 200 us enabled phase 0, inhibit for 50 us, 200 us phase 1, inhibit.

No MCU, ADC, current monitor, patient isolation, implant package or validated safety interlock populated.

R8/R9/R10: 5*(1.02k)/(49.9k+1.02k)/1k = 100.157 uA nominal; calibrate if ever fabricated.

Digital reference only; no body connection

Project ATO

Sheet: /
File: ato_reference.kicad_sch

Title: ATO | Artificial−signal analog reference

Size: A3	Date:	**Rev: 1.0**
KiCad E.D.A. 10.0.6		Id: 1/1